Review Report

Environmental Research Letters

Manuscript: Tropical Cyclone Wind Intensity Drives Phase-Specific Displacement of Emergency Department Demand in Florida, U.S., 2004–2022 (ERL-125149)

Summary

The manuscript addresses a genuine gap in the tropical cyclone (TC) health literature: most prior studies anchor exposure windows at or after landfall, leaving the pre-impact ("Threat") period under-characterized. Using 19 years of ZIP-code-level emergency department (ED) data from Florida and a two-part fixed-effects model, the authors identify a consistent Threat-increase / Impact-decrease / Cleanup-rebound pattern that scales systematically with storm intensity, supported by two well-designed falsification analyses (timing randomization and label permutation). This is a credible and novel core finding, and the falsification work is a genuine strength of the paper. However, several methodological details require clarification before the results can be fully trusted, and the causal framing, particularly in the title, should be softened given the observational design.

Major Comments

1. Storm swath definition needs clarification (Section 2.2). The text states swaths were built from IBTrACS "based on the radius of maximum wind." RMW is typically a tight radius near the eyewall, not the areal extent of ≥34/≥55/≥64 kt winds. If exposure classification is actually based on radius of gale-force/storm-force wind (R34/R50/R64 wind radii fields) rather than literal RMW, this is likely a wording issue, but as written it implies ZIP-code exposure is defined by proximity to the storm's tightest wind core, which would dramatically undercount exposed ZIP-days for large-diameter storms and bias the whole exposure metric. This needs an explicit, unambiguous definition (which wind radii field, what interpolation between synoptic-time IBTrACS fixes, how coastal versus inland ZIPs are assigned).

2. Suppression handling loses resolution in the range that matters most for the paper's own explanation of its results. ZIP-days with true counts ≤6 are all coded as "6," and this suppression rule applies uniformly to all ZIP-days regardless of location. However, it binds disproportionately often in lower-population ZIPs, which tend to be rural, simply because these ZIPs are more likely to have naturally low daily counts. Since the Discussion attributes the strong intensity-scaling pattern specifically to higher-volume urban/suburban ZIP codes, the paper should show how estimates differ by population density strata, or at least discuss how suppression-driven information loss in lower-volume ZIP-days might affect the robustness or generalizability of the reported pattern, particularly the "modest and inconsistent" Threat effect.

3. Study period spans COVID-19 (2020–2022), when ED utilization fell sharply nationwide for reasons unrelated to storms. Year-month fixed effects should absorb secular/seasonal shifts, but a structural shock of COVID's magnitude interacting with storm-year composition (how many of the 44 qualifying storms fall in 2020–2022?) deserves explicit treatment — a sensitivity analysis excluding 2020–2022, or at least a sentence confirming results are stable with/without those years.

4. Overlapping storms are dismissed as "rare" without support. The manuscript's own cited reference [2] confirms that four hurricanes (Charley, Frances, Ivan, and Jeanne) struck Florida within the 2004 season alone. Given this, a ZIP code could plausibly be in Cleanup from one storm while entering Threat for the next within the same season. The stated priority rule (Impact > Threat > Cleanup) resolves same-storm ambiguity but doesn't obviously handle cross-storm sequencing — e.g., is a ZIP-day in Cleanup(A) and Threat(B) coded as Threat, or dropped, or does one

override entirely? This should be stated explicitly and, ideally, quantified (how many ZIP-days are affected) rather than asserted as rare.

5. Two-part model combination method is under-specified. Section 2.3 says coefficients were "translated into marginal, phase-specific expected ED visits" by counterfactually reassigning phase labels and "averaging predictions from the two-part model." This is presumably a recycled-predictions/g-computation approach (predict under each counterfactual phase for every ZIP-month, average, weight by population), but the exact formula combining the logistic and Poisson components into a single expected-count estimate isn't given. Since this combination is the paper's central quantity, it needs an explicit equation, not just a prose description.

6. Causal language exceeds what the design supports. The title ("Wind Intensity Drives...") and phrases like "drives phase-specific displacement" assert causality. The two placebo/falsification analyses are a genuine strength and meaningfully raise confidence, but this remains an observational, non-randomized design with fixed effects rather than a natural experiment with truly exogenous variation in storm timing and exposure. I'd recommend "is associated with" throughout, including the title.

Minor Comments

- Population interpolation (Section 2.4): "interpolated available years to a complete annual series" and "carried the nearest available estimate for years outside the ACS range" needs more detail — linear interpolation? Which years fall outside the ACS 5-year range at the start of the study period (2004–2008, before ACS 5-year estimates existed), and how much of the population series relies on this extrapolation?
- Bootstrap CIs: described as "200 ZIP-clustered bootstrap resamples" — please confirm whether this also blocks on time (e.g., resampling contiguous day-blocks or storms as a unit) to account for day-to-day serial correlation within a ZIP, or whether resampling clusters only on ZIP, which could understate uncertainty given that storm-phase days are consecutive by construction.
- A summary table with point estimates and CIs for all phases × intensity thresholds (currently presented only in Figure 1 and in prose) would improve readability and citability.
- Section 3.2 reports nearly a dozen point estimates and CIs in-line in prose, duplicating what Figure 1 already shows visually. Consider moving detailed values to a summary table and simplifying the prose to describe the pattern (e.g., monotonic scaling direction and approximate magnitude), reserving exact figures for the table/figure only.
- Terminology: "≥55 kt" is described as a "stronger tropical-storm threshold" (line 119). I recommend using the numeric thresholds consistently throughout rather than introducing an additional descriptive label, to avoid ambiguity with Saffir-Simpson categories.
- Self-citation [9]: the companion mortality paper is listed as "In Review." Please confirm current status (in review, accepted, or published) before final submission and update the reference accordingly.
- Data/code availability statement: "synthetic examples will be shared as permitted" is vague. A rough timeline or contact process would strengthen the availability statement.
- Practical significance: +0.32 and −1.45 visits per ZIP-day against a baseline of 17.5 are roughly 2% and 8% shifts respectively. Worth a sentence in the Discussion translating this into absolute surge-capacity terms (e.g., visits per 100,000 population per storm) for the stated goal of informing staffing decisions.
- Reference/citation order: several multi-reference in-text citations are not in ascending numerical order, e.g. "[6, 15, 3]" (line 207), "[16, 3, 17]" (line 232), and "[6, 1]" (line 217). While not an explicit ERL rule,

ascending order within a bracketed citation is the standard convention for numbered reference systems and would improve readability. Recommend reordering to "[3, 6, 15]", "[3, 16, 17]", and "[1, 6]" respectively.

Recommendation

Major Revision.

The core contribution, a phase-resolved, ZIP-code-level characterization of ED utilization around TC landfall, with intensity scaling and rigorous falsification testing, is a genuine addition to the literature, and I do not think this warrants rejection. However, the exposure-definition ambiguity (Comment 1), the suppression/resolution issue interacting with the paper's own explanatory claims (Comment 2), the unaddressed COVID-period confound (Comment 3), the unquantified overlapping-storm handling (Comment 4), and the unspecified model-combination formula (Comment 5) all bear directly on whether the reported effect sizes can be taken at face value. These should be resolved, and where relevant re-verified with additional analysis, before the manuscript is reconsidered.